

Cross-Platform Mobile App Development

Week 4 — Hybrid Mobile with Capacitor

Instructor: Nguyen Thanh Tuan, PhD

Institution: Faculty of Computer Science, VKU

Today's Agenda

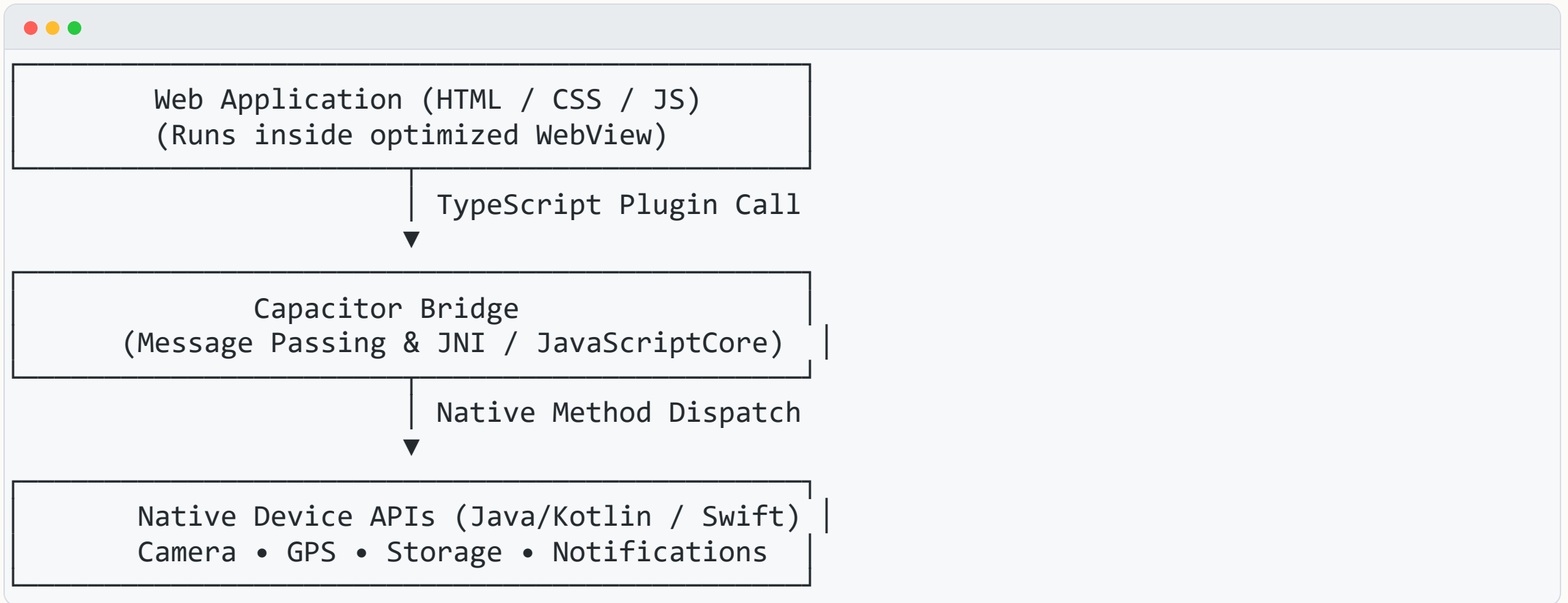
1. The Hybrid Model: WebView + Native Bridge
2. Why Capacitor replaced Apache Cordova / PhoneGap
3. Capacitor runtime architecture & JavaScript-to-Native IPC
4. Accessing device hardware: `@capacitor/camera` , `@capacitor/geolocation` ,
`@capacitor/network`
5. Local notifications & secure preferences storage
6. Packaging your Week 3 PWA into an installable Android APK
7. Wrap-up & Mini-Project 1 Submission Checklist

Part 1: Hybrid Architecture & Capacitor

Evolution: Cordova vs. Capacitor

Aspect	Apache Cordova	Capacitor (Modern Hybrid)
Architecture	Black-box build scripts	Native iOS/Android projects as source artifacts
Tooling	Custom CLI wrappers	Native IDEs (Android Studio & Xcode)
Plugin API	Global callbacks (<code>window.plugins</code>)	Modern Promise/async-await TypeScript APIs
PWA Support	Non-existent	First-class citizen — same code runs as PWA and Native
Maintenance	Deprecated / Legacy	Actively maintained by Ionic team

How the Capacitor Bridge Works



Part 2: Essential Native Plugins

Core Capacitor Plugins

```
• • •  
# Camera capture & photo gallery  
npm install @capacitor/camera  
  
# Real-time network status  
npm install @capacitor/network  
  
# Device GPS Location  
npm install @capacitor/geolocation  
  
# Local in-app notifications  
npm install @capacitor/local-notifications
```

Practical Example: Camera Capture

```
import { Camera, CameraResultType, CameraSource } from '@capacitor/camera';

async function takeFacilityPhoto() {
  const image = await Camera.getPhoto({
    quality: 85,
    allowEditing: false,
    resultType: CameraResultType.Base64,
    source: CameraSource.Camera
  });

  return `data:image/jpeg;base64,${image.base64String}`;
}
```

Part 3: Mini-Project 1 Submission

Mini-Project 1 Deliverables Checklist

1.  **Live Demo URL:** Live deployed PWA on Cloudflare Pages / Vercel with HTTPS.
2.  **GitHub Repository:** Public repository containing both PWA source and `capacitor.config.ts`.
3.  **Short Technical Report (PDF):** 2–4 pages adhering to the official template with annotated screenshots.
4. **Deadline:** End of Week 4 (Sunday 23:59).

Next Week: React Native & Expo

- In Week 5, we begin **React Native** — moving from WebViews to true Native Components!